

Fabio Cuccaro

Naples, Italy

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Personal Profile

A passionate and enthusiastic Automation and Robotic Engineer, with an intense thirst for knowledge, fascinated by innovations and discoveries. My will to make a positive impact on the world, as well as share and learn from others, highly motivates me to tackle new challenges. With experience in Model-based Control for the automotive sector, my interests lie in creating solutions to control Robotic engineering challenges.

Education

University of Naples Federico II

Naples, Italy

MSc in Automation and Robotic Engineering

Mar2020 - Sep 2025

- **Thesis:** Design and Experimental Evaluation of a ROS-Based Social Robot for Multimodal Mood and Health Monitoring in Office Environments. This thesis was developed at the Tohoku University of Japan, working on social robotics, computer vision, autonomous navigation of robot and Human Robot Interaction
- **Courses:** Computer-Aided Design, Biomedical and Advanced Robotics, Linear and Non-Linear Control Theory and Methods, Optimal Control, Real-Time Systems, Control of Electrical Machines, Mobile Robotics

University of Naples Federico II

Naples, Italy

BSc in Automation Engineering

Sept 2016 - Jan 2020

- **Thesis:** Reduced order models for vertical stabilization in the Tokamak EAST
- **Courses:** Automatic Control Theory and Methods, Automation Technology, PLC, Electrical Machines, Electronics Fundamentals, Signals Theory and Mechanical Fundamentals

Experience

Smart Robot Design Lab, Tohoku University

Sendai, Japan

Research Intern

Mar 2024 - Oct 2024

- Worked on the "Sunny" social robot project for my master's thesis.
- Developed functionalities for multi-modal mood detection using Facial Expression Recognition (FER) and Speech Emotion Recognition (SER).
- Implemented health monitoring features using a thermal camera and mmWave sensor to measure the heart rate of the interlocutor.
- Managed the data communication system organized via ROS communication and integrated the detection module with the interaction and navigation modules of the robot.
- The goal of Sunny is to detect the mood of the interlocutor and attempt to positively impact their mood during the interaction.
- **Technical Skills:** C++, Python, ROS, OpenCV, TensorFlow, PyTorch, pretrained AI models, GitHub.
- **Soft Skills:** Research and Development, Problem-solving, Time Management, Team Collaboration, Technical Documentation.

University Projects

Impedance Controller for a LBR KUKA

Naples, Italy

University of Naples

Dic 2023 - Gen 2024

- Implemented on ROS an Impedance Controller algorithms for a simulated 7-DOF LBR iiwa KUKA collaborative robot for medical applications.
- **Technical Skills:** ROS (Gazebo, KDL), C++, Overleaf, LaTeX, Matlab.
- **Soft Skills:** Time Management, Presentation skills, Report writing.

Multirobot system with Turtlebot and Kuka Iiwa

Naples, Italy

University of Naples

Sep 2023 - Nov 2023

- The Turtlebot autonomously maps and navigates the environment using SLAM (Simultaneous Localization and Mapping) techniques. OpenCV libraries were used for image analysis and QR code detection. Precise position control of the Kuka iiwa robotic arm was implemented with ROS CNTRL.
- **Technical Skills:** ROS (Gazebo, KDL), C++, Overleaf, LaTeX, Matlab.
- **Soft Skills:** Time Management, Presentation skills, Report writing.

Matlab-based Control and Analysis of a SCARA Manipulator

Naples, Italy

University of Naples

Mar 2023 - Apr 2023

- Use of Matlab/Simulink environment to develop algorithms for kinematic inversion with inverse, pseudo-inverse and transposed Jacobian and the design of adaptive, robust and inverse dynamic controls
- **Technical Skills:** Matlab, Simulink, C++, Overleaf, LaTeX.
- **Soft Skills:** Time Management, Presentation skills, Report writing, Problem Solving.

- Collaborated with a four-person team for the development of a firmware onboard of an electric vehicle, aimed at handling different protocols (CANBus, SPI, USART) needed for the communication between the STM32 board and the various devices (BMS, Inverter, Sensors).
- Use of Matlab/Simulink environment for the realization of the high-level code aimed at handling the various safety procedures on board (emergency shutdown, BMS/Inverter Thermal Management, Plausibility Check on signals, Torque Vectoring Control) as well as the implementation of algorithms to realize ABS and Traction Control
- Tasks Organization needed for the accomplishment of the imposed objectives and the cooperation with the Electronic and Mechanical Teams
- **Technical Skills:** Matlab/Simulink and Tools, C++/#, Git, STM32, Soldering, Electric Circuit, FreeRTOS.
- **Soft Skills:** Teamwork, Time and Tasks Management, Communication, Requirements Handling, Problem Solving, Presentation skills, Report writing, Logical and Critical Thinking.

Skills

Programming ROS, Matlab/Simulink (Control, Optimization, State Chart, CodeGen), RTOS (FreeRTOS, Chibios, RTAI), C/C++, Python.

Miscellaneous Catia, Linux, $\text{\LaTeX}$ (Overleaf), Microsoft Office, Git.

Soft Skills Time Management, Teamwork, Problem-solving, Documentation and Data Interpretation, Critical and Logical Thinking.

Languages

English Full Professional proficiency, C1 certified by the university language office (CLA) placement test

Japanese Basic proficiency, completed basic Japanese course at Tohoku University

Italian Native proficiency